

Hands-on 4: Difference between JPA, Hibernate and Spring Data JPA

Java Persistence API (JPA)

- JPA (Java Persistence API) is a **specification** for managing data between Java objects and relational databases.
- It defines standards for Object Relational Mapping (ORM).
- JPA **does not provide an implementation**.
- It requires an implementation such as Hibernate.

Hibernate

- Hibernate is an **ORM framework**.
- It is one of the **implementations of JPA**.
- It provides features such as:
 - CRUD operations
 - Transaction management
 - Mapping Java classes to database tables
- In Hibernate, developers manually manage sessions and transactions.

Example

```
Session session = factory.openSession();  
  
Transaction tx = session.beginTransaction();  
  
session.save(employee);  
  
tx.commit();  
  
session.close();
```

Spring Data JPA

- Spring Data JPA is a **Spring framework module** built on top of JPA.
- It does **not implement JPA**.

- It reduces boilerplate code.
- It automatically manages transactions using @Transactional.
- CRUD operations are performed using repository interfaces.

Example

@Autowired

```
private EmployeeRepository employeeRepository;
```

@Transactional

```
public void addEmployee(Employee employee) {
    employeeRepository.save(employee);
}
```

Comparison Table

Feature	JPA	Hibernate	Spring Data JPA
Type	Specification	JPA Implementation	Spring Framework
Implementation	No	Yes	No
ORM Support	Defines ORM rules	Provides ORM implementation	Uses Hibernate/JPA
Boilerplate Code	More	Moderate	Very Less
Transaction Management	No	Manual	Automatic
CRUD Operations	Using EntityManager	Using Session	Using Repository
Ease of Use	Moderate	Easy	Easiest

Relationship

Spring Data JPA



Hibernate



JPA



Database

Conclusion

- **JPA** is a specification that defines how Java objects should be persisted.
- **Hibernate** is the most popular implementation of JPA and provides ORM functionality.
- **Spring Data JPA** is built on top of JPA and Hibernate and minimizes boilerplate code by providing repository interfaces and automatic transaction management.